

Credit Name: CSE2920 - CSE Project C

Assignment Name: Build A Thermostat

How has your program changed from planning to coding to now? Please explain?

PLANNING:

I plan to create a code that will start with a set temperature of 21 degrees C, the red button decreases the set temp by 1, and the green button increases the set temp by 1. The current temp and set temp will print every 10 seconds, if the current temp is 2 degrees C within the set temp, the green LED will stay on, otherwise if it is outside that range, the red LED will turn on.

CODING:

1.

```
import com.phidget22.DigitalInput;

public class BuildThermostat {

    public static void main(String[] args) throws Exception{

        //Create | Create objects for your buttons and LEDs.
        DigitalInput redButton = new DigitalInput();
        DigitalOutput redLED = new DigitalOutput();
        DigitalInput greenButton = new DigitalInput();
        DigitalOutput greenLED = new DigitalOutput();
        TemperatureSensor temperatureSensor = new TemperatureSensor();

        //Address | Address your four objects which lets your program k
        redButton.setHubPort(0);
        redButton.setIsHubPortDevice(true);
        redLED.setHubPort(1);
        redLED.setIsHubPortDevice(true);
        greenButton.setHubPort(5);
        greenButton.setIsHubPortDevice(true);
        greenLED.setHubPort(4);
        greenLED.setIsHubPortDevice(true);

        //Open | Connect your program to your physical devices.
        redButton.open(1000);
        redLED.open(1000);
        greenButton.open(1000);
        greenLED.open(1000);
        temperatureSensor.open(1000);
```

Phidgets library is imported

temperatureSensor, redButton, redLED, greenButton, and greenLED objects are created to interact with the physical phidgets box.

The HubPort number was set based on the appropriate object.

IsHubPortDevice set to true for all objects (confirming to the program that the objects are part of phidgets box).

Opened all objects, establishing a connection to the physical Phidgets box.

2.

```
//temp that user sets. default temp is 21C
int SetTemp = 21;

//current temperature variable
double CurrentTemp = temperatureSensor.getTemperature();

while (true) {
    // 100 iterations, each with 100ms delay = 10 seconds
    for (int i = 0; i < 100; i++) {
```

SetTemp variable to store the temperature set by the user.

The current temp variable gets the actual temperature of the room.

While loop is used so that the following code repeats.

For loop with 100 iterations, and it has a 100ms delay near the end of the for loop code. 100 iterations and 100ms = 10 seconds before the Current temp and SetTemp both print.

I used this for loop because without it, I wasn't able to actively change the temperature. I had to wait 10 seconds before being able to change the temperature. With the for loop, I can set the temperature almost instantly, while still printing the Current temp and SetTemp every 10 seconds.

3.

```
//green button increases SetTemp, red button decreases
if (greenButton.getState()) {
    SetTemp++;
    System.out.println("The Set temperature increased to: " + SetTemp + " °C");
    while (greenButton.getState()) {
        Thread.sleep(10);
    }
}

if (redButton.getState()) {
    SetTemp--;
    System.out.println("The Set temperature decreased to: " + SetTemp + " °C");
    while (redButton.getState()) {
        Thread.sleep(10);
    }
}
```

If the green button is pressed, a value of 1 is added to SetTemp, and program prints that the temperature was increased and then outputs the SetTemp.

While the green button is pressed, there is a sleep time of 10ms to eliminate any invalid button presses.

If the red button is pressed, a value of 1 is removed to SetTemp, and program prints that the temperature was decreased and then outputs the SetTemp.

While the green button is pressed, there is a sleep time of 10ms to eliminate any invalid button presses.

4.

```
// if the absolute values of the current temp and set temp are close enough, then the green LED will turn on.
//else red led will turn on.
if (Math.abs(CurrentTemp - SetTemp) <= 2) {
    redLED.setState(false);
    greenLED.setState(true);
}
else {
    redLED.setState(true);
    greenLED.setState(false);
}
Thread.sleep(100); //100ms delay for each iteration
}
```

If the absolute value of the current temp minus the set temp is less than or equal to 2, then the green LED will stay on. Else, the red LED will stay on.

The Sleep time of 100ms for the for loop that I discussed in step 2 is at the end of the for loop. This is where the for loop ends.

5.

```
// print the set temp and current temp every 10 seconds.
System.out.println("Current Temperature: " + CurrentTemp + " °C");
System.out.println("The Set temperature is: " + SetTemp + " °C");

}
```

The program outputs the current temp along with the set temp in degrees Celsius. It prints every 10 seconds, due to the for loop that I placed previously with 100 iterations of 100ms, which equates to 10 seconds.

End Of Program!